

Flow Over Cylinder: **Elastic Material Simulation**

Set up details:

Physics Interface: Fluid–Solid Interaction (FSI)

- Laminar Flow
- Solid Mechanics
- Moving Mesh (ALE)

Study Type: Time-Dependent Analysis

Boundary Conditions:

- Inlet: Fully developed laminar flow with time-dependent velocity profile
- Outlet: Pressure = 0 Pa
- Channel Walls: No-slip condition
- Filament Base: Fixed constraint

Mesh:

- Free Triangular Mesh
- Mesh Size: Fine
- Calibrated for Fluid Dynamics

Outputs Evaluated:

- Velocity distribution
- Pressure distribution
- Stress distribution
- Filament deformation
- Mesh velocity
- Flow animation

Geometry

Parameter	Previous Model	Tutorial Model
Channel Height	30 m	100 μm
Channel Width	37 m	300 μm
Filament Width	1 m	5 μm
Filament Height	14 m	47.5 μm
Filament Position (x)	12 m	100 μm
Fillet Radius	0.5 m	2.5 μm

Material

Surrounding Fluid Material

Property	Previous Model	Tutorial Model
Material	Air	Blank Material
Density	1.22 kg/m ³	1000 kg/m ³
Dynamic Viscosity	0.0122 Pa·s	0.001 Pa·s

Elastic Filament Material

Property	Previous Model	Tutorial Model
Material	Blank Material	Blank Material
Young's Modulus	1×10 ⁵ Pa	2×10 ⁵ Pa
Poisson's Ratio	0.49	0.33
Density	100 kg/m ³	7850 kg/m ³

Inlet Boundary Condition

Parameter	Previous Model	Tutorial Model
Inlet Type	Normal Inflow Velocity	Fully Developed Flow
Velocity	1 m/s / 0.1m/s	$U \cdot \sin^2(t)$
Steady-State Velocity	1 m/s	3.33 cm/s

Solid Mechanics

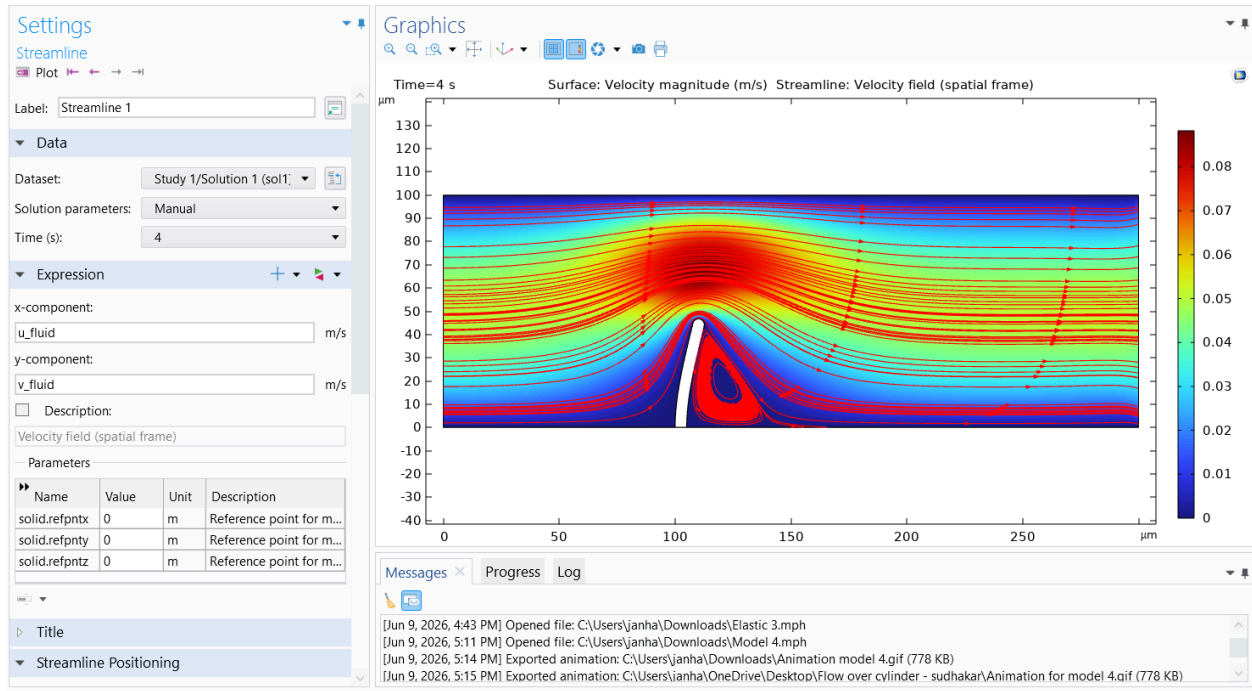
Property	Previous Model	Tutorial Model
Young's Modulus	1×10^5 Pa	2×10^5 Pa
Poisson's Ratio	0.49	0.33
Density	100 kg/m ³	7850 kg/m ³

Time Dependent Study

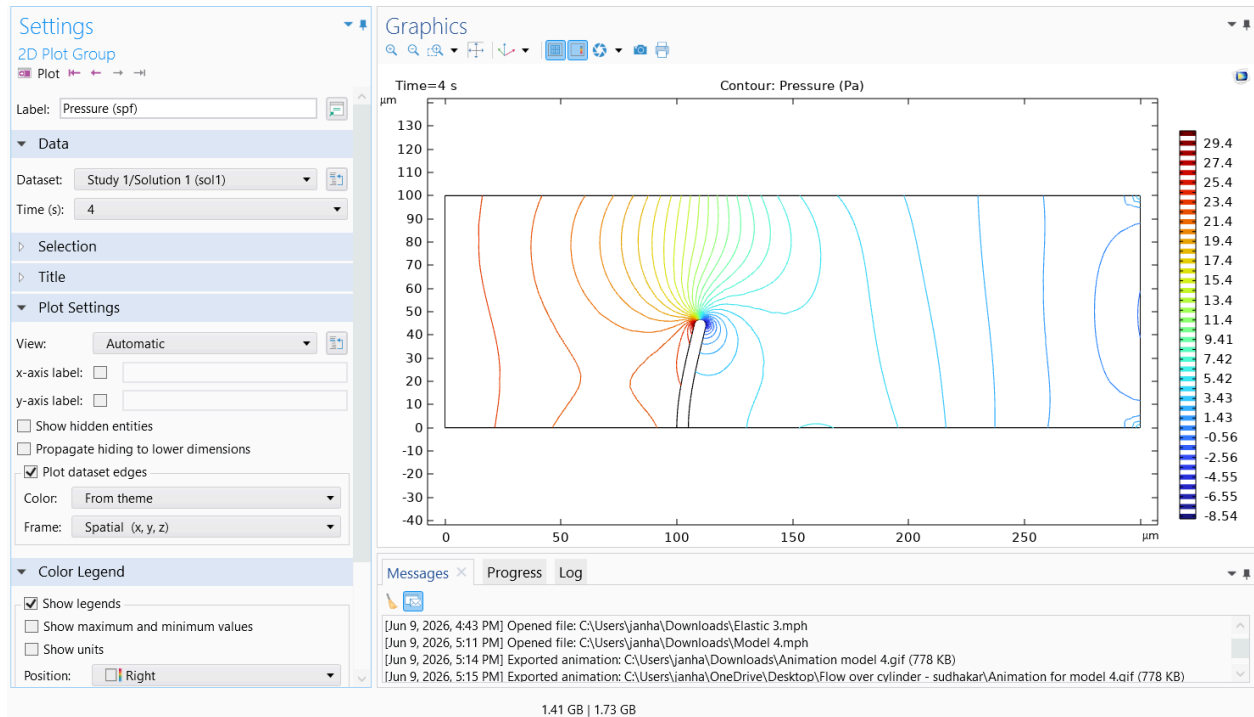
Parameter	Previous Model	Tutorial Model
Output Times	range(0,0.001,7)	range(0,0.005,0.75), range(1,0.25,4)
Relative Tolerance	1×10^{-6}	1×10^{-4}

Results:

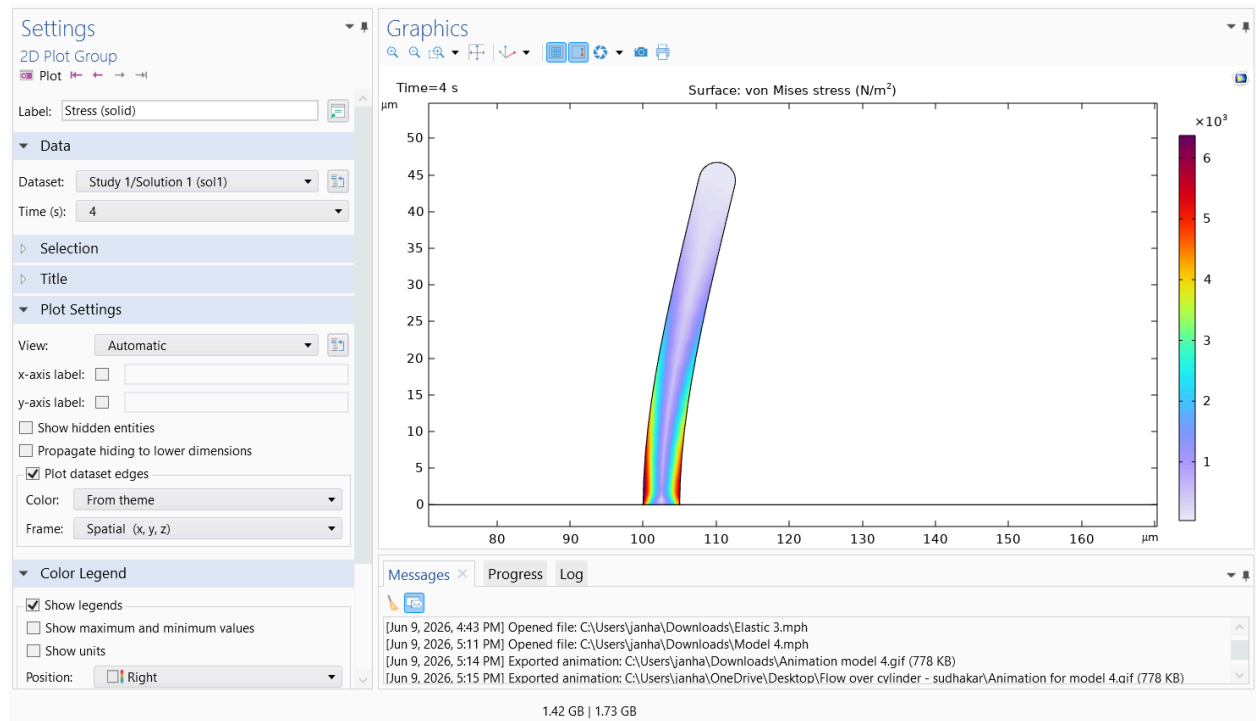
1. Velocity Plot at t=4



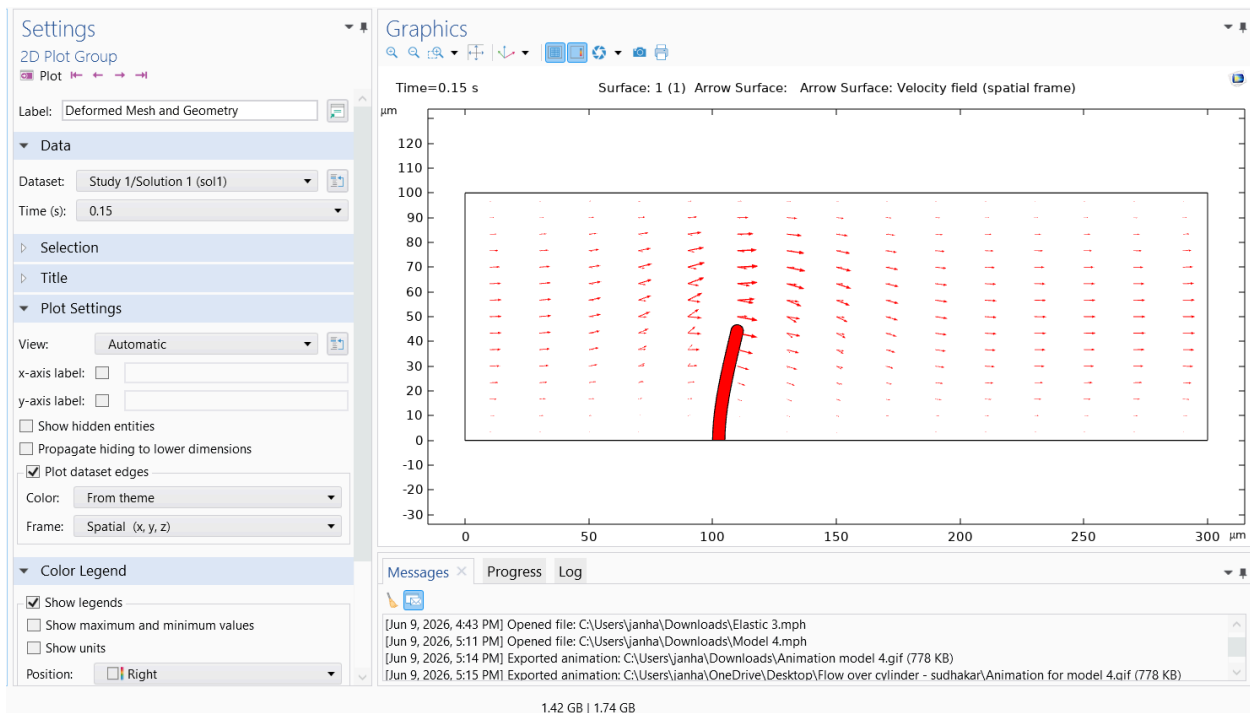
2. Pressure Plot :



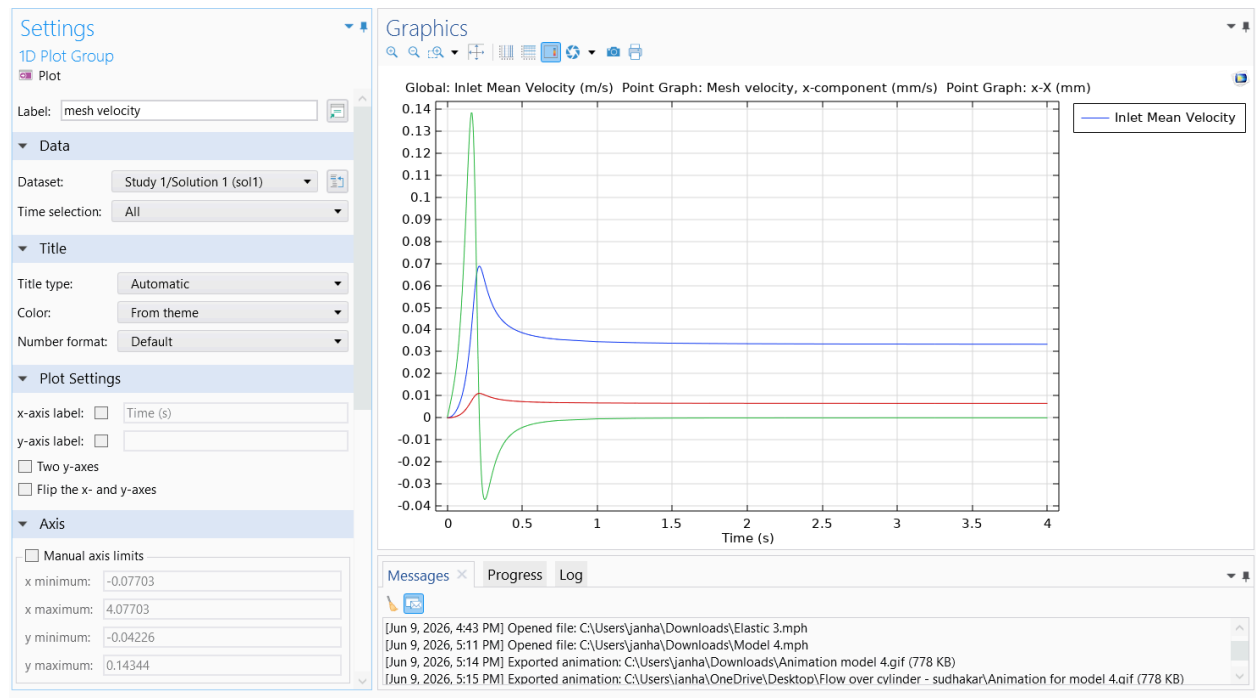
3. Stress:



4. Deformed Mesh and Geometry:



5. Mesh Velocity:



Animation:

